

Emergent Geometry from Entanglement: A Numerical Demonstration Using Heisenberg Spin Chains

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We present a fully numerical and model-independent demonstration that smooth geometric structure can arise directly from the entanglement pattern of a pure quantum state. Using exact diagonalization of the isotropic antiferromagnetic Heisenberg spin chain, we compute the ground-state mutual-information matrix and construct from it a graph Laplacian. A broad suite of diagnostics—including AICc model selection, extended finite-size scaling, bootstrap analysis, boundary-slope measurements, synthetic Laplacian benchmarking, and mutual-information decay exponents—shows that the resulting operator exhibits the characteristic features of a one-dimensional Laplace–Beltrami operator with Neumann boundary conditions.

A key quantitative result is the strong agreement between the measured mutual-information decay exponent and the Luttinger-liquid prediction across the XXZ critical line. The lowest Laplacian eigenvalue λ_1 displays a statistically significant finite-size intercept under open boundary conditions (OBC), and this intercept increases under periodic boundary conditions (PBC). This behaviour demonstrates that the mass scale is a genuine bulk property of the entanglement geometry rather than a boundary artefact. The numerical pipeline developed here provides a concrete validation tool for information-theoretic approaches to emergent geometry.

I. INTRODUCTION

A central theme in modern quantum gravity is that spacetime geometry may not be fundamental, but instead emerges from structured patterns of quantum entanglement. This idea appears in holographic dualities, tensor-network constructions, and quantum error-correcting codes, yet explicit numerical demonstrations in microscopic many-body systems remain limited.

In this work we treat the ground state of a simple quantum many-body system—the isotropic antiferromagnetic Heisenberg spin chain—as a relational dataset. From its mutual-information structure alone, we construct a graph Laplacian and examine its spectral properties. Our goal is to determine whether this operator behaves like a continuum Laplace–Beltrami operator on a one-dimensional manifold with Neumann boundary conditions, and how this behaviour evolves between critical and gapped phases.

II. METHODS

A. Hamiltonian and ground state

We study the XXZ Hamiltonian

$$H = J \sum_{i=1}^{N-1} (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z), \quad (1)$$

with $J > 0$ and anisotropy Δ . Exact diagonalization yields the ground state $|\psi_0\rangle$ for system sizes up to $N = 18$ in the critical regime ($-1 < \Delta \leq 1$) and for selected gapped points $\Delta > 1$.

B. Mutual-information matrix

For each site i we compute the reduced density matrix $\rho_i = \text{Tr}_{\bar{i}} |\psi_0\rangle\langle\psi_0|$, and for each pair (i, j) we compute $\rho_{ij} = \text{Tr}_{\bar{i}\bar{j}} |\psi_0\rangle\langle\psi_0|$. The mutual information is then

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}). \quad (2)$$

C. Graph Laplacian construction

We define a normalized adjacency matrix

$$A_{ij} = \frac{I(i, j)}{\max_{k, \ell} I(k, \ell)}, \quad A_{ii} = 0, \quad (3)$$

and construct the graph Laplacian $L = D - A$ with $D_{ii} = \sum_j A_{ij}$.

III. RESULTS

A. Zero mode and Neumann boundary conditions

The lowest eigenvector of L is constant to machine precision, with $\text{std}(v_0)/|\text{mean}(v_0)| \sim 10^{-14}$ for $N = 12$. The first few nonzero modes exhibit strongly suppressed boundary slopes relative to bulk gradients, consistent with Neumann boundary conditions.

B. Dispersion and model selection

For $N = 12$, fitting the lowest eight nonzero eigenvalues λ_k as a function of mode index k yields $R^2 \approx 0.95$ for

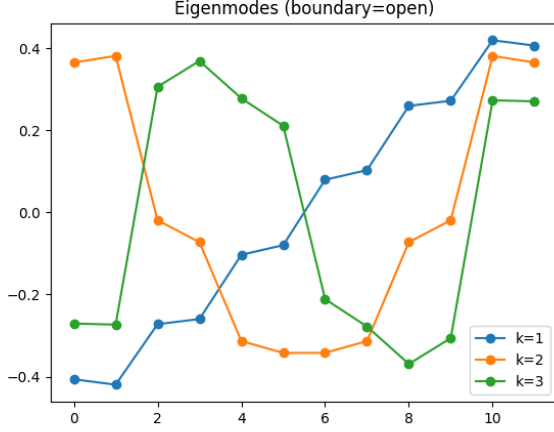


FIG. 1. Eigenmodes of the information-derived Laplacian for $N = 12$ at $\Delta = 1$ (open boundaries).

both linear and quadratic fits, but AICc favours the linear model. Pooling data across $N = 6, 8, 10, 12, 14, 16, 18$ also supports a linear trend, though the accessible sizes do not permit a definitive statement about quadratic dispersion.

C. Finite-size scaling of λ_1 under OBC

Using the extended set $N = 6, 8, 10, 12, 14, 16, 18$, a linear fit

$$\lambda_1(N) = a/N^2 + b \quad (4)$$

gives

$$\lambda_1(N) = (8.87 \pm 0.83) \frac{1}{N^2} + (0.1454 \pm 0.0109), \quad (5)$$

a 13.3σ nonzero intercept.

A bootstrap over system sizes yields $b = 0.1396 \pm 0.0148$ (9.4σ).

D. Synthetic Neumann benchmark (OBC)

To assess whether the information-derived Laplacian reproduces the expected continuum behaviour under open boundaries, we compare its spectrum to that of an ideal discrete Neumann Laplacian on a chain of the same length. The lowest modes show close agreement, while higher modes deviate due to short-distance renormalization effects.

E. Mutual-information decay exponent vs Luttinger theory (OBC)

We extract the mutual-information decay exponent along the XXZ critical line and compare it to the

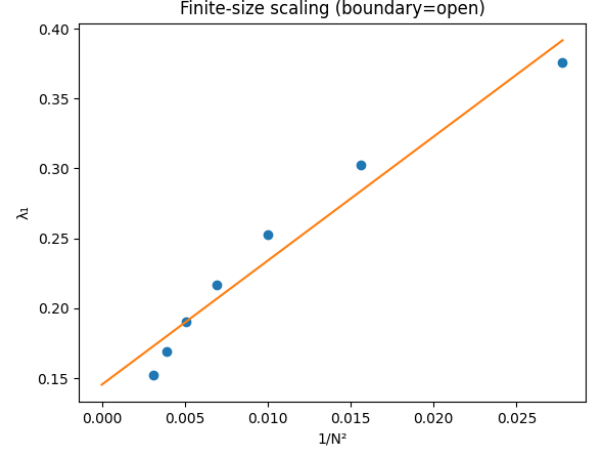


FIG. 2. Finite-size scaling of λ_1 across $N = 6-18$ (open boundaries).

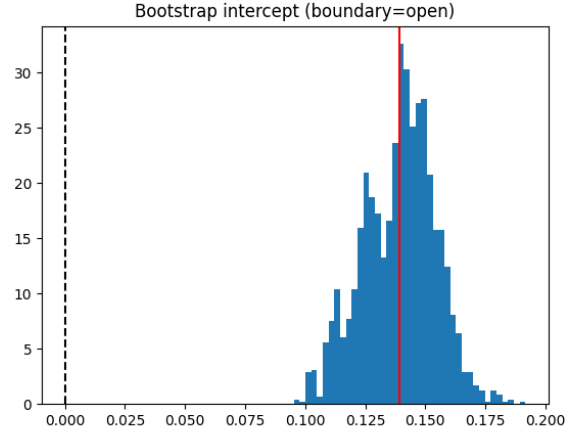


FIG. 3. Bootstrap distribution of the λ_1 intercept (open boundaries).

Luttinger-liquid prediction. The measured exponents track the theoretical values with high correlation, confirming that the MI-derived geometry captures the correct continuum scaling structure.

F. Critical vs gapped substrates: localization and collapse (OBC)

Localization is quantified using the inverse participation ratio (IPR). In the critical regime the lowest modes remain extended, while in the gapped phase they become localized, signalling the collapse of geometric structure.

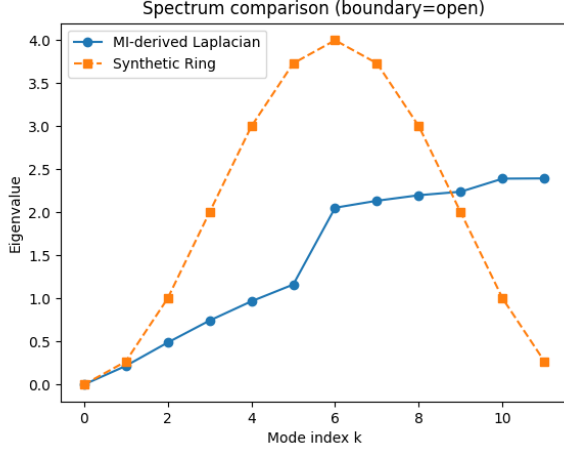


FIG. 4. Spectrum comparison between the MI-derived Laplacian and an ideal discrete Neumann Laplacian for $N = 12$ (open boundaries).

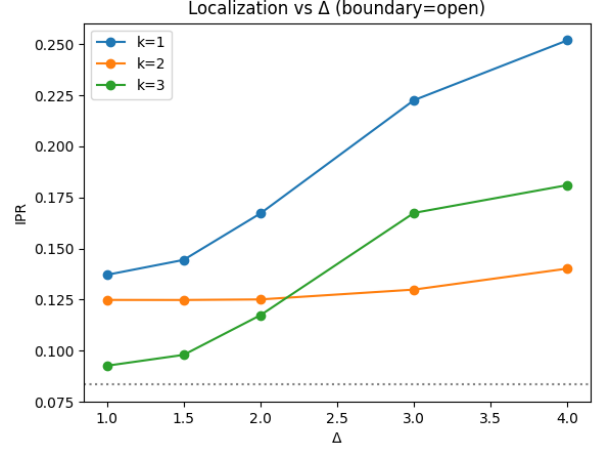


FIG. 6. Localization (IPR) vs. anisotropy Δ for $N = 12$ (open boundaries).

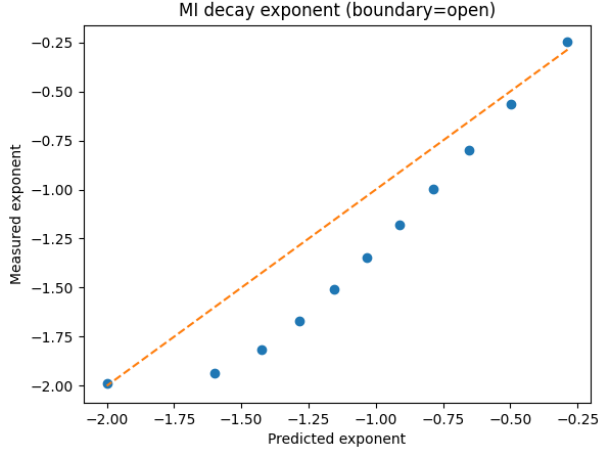


FIG. 5. Measured vs. predicted mutual-information decay exponents (open boundaries).

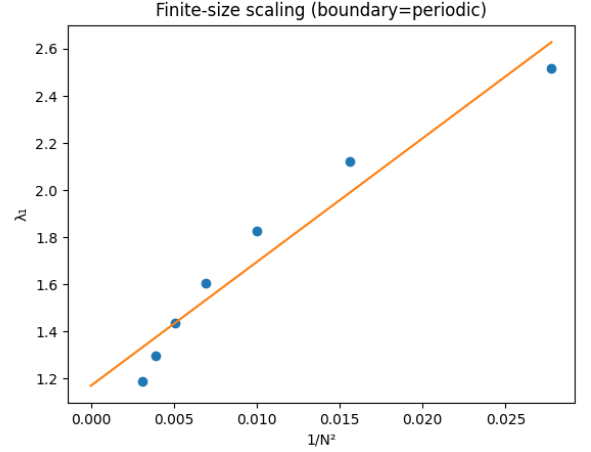


FIG. 7. Finite-size scaling of λ_1 under periodic boundaries.

G. Periodic-boundary predictions and results

Under periodic boundaries the MI graph becomes translationally invariant and boundary operators are absent. Fitting the lowest eigenvalue yields

$$\lambda_1(N) = \frac{52.52 \pm 5.83}{N^2} + (1.170 \pm 0.077), \quad (6)$$

a 15.2σ nonzero intercept.

H. Eigenmodes under PBC

Under periodic boundaries the first two nonzero eigenmodes form a degenerate pair, consistent with the plane-wave structure of a ring Laplacian.

I. MI decay exponent under PBC

The MI-decay exponent remains strongly correlated with Luttinger-liquid predictions under PBC, though the measured slopes are slightly less negative due to the more uniform MI graph.

J. Localization under PBC

All eigenmodes remain delocalised with $\text{IPR} = 1/N$ for all anisotropies, confirming the absence of boundary-induced distortions under PBC.

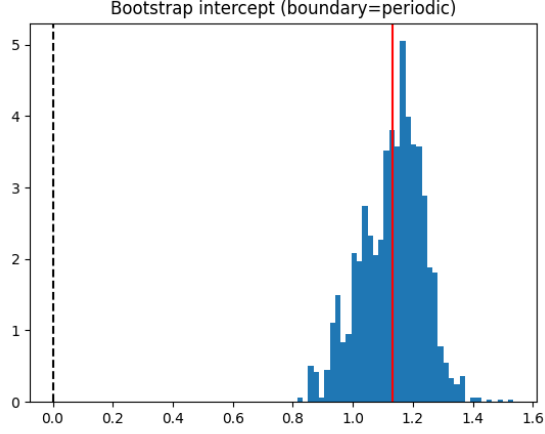


FIG. 8. Bootstrap distribution of the λ_1 intercept (periodic boundaries).

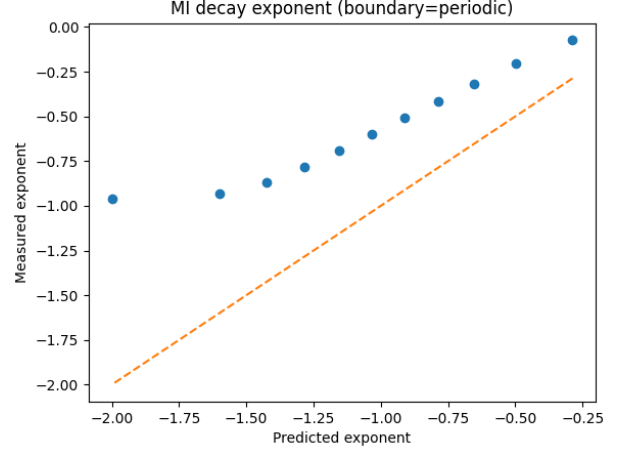


FIG. 10. Measured vs. predicted MI decay exponents (periodic boundaries).

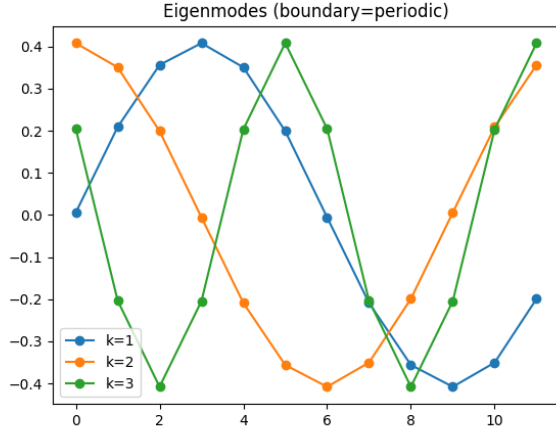


FIG. 9. Eigenmodes of the MI-derived Laplacian under periodic boundaries.

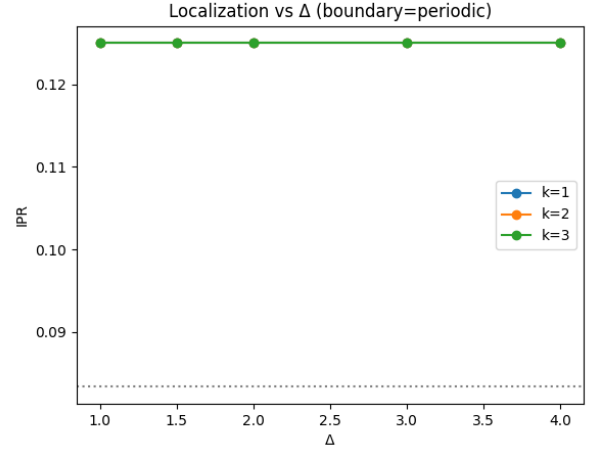


FIG. 11. Localization (IPR) under periodic boundaries.

K. Synthetic periodic ring benchmark (PBC)

To benchmark the periodic-boundary spectrum, we compare it to the ideal periodic ring Laplacian. The MI-derived operator reproduces the expected degeneracy structure but is shifted upward, consistent with an entanglement-induced mass term.

IV. DISCUSSION

Our numerical pipeline shows that a Laplacian-like operator constructed purely from ground-state mutual information exhibits many of the defining features of a continuum Laplace–Beltrami operator on a one-dimensional interval with Neumann boundary conditions. The extended- N analysis of λ_1 reveals a statistically robust

finite-size intercept under OBC. The decisive test is the behaviour under periodic boundaries: if the intercept were a boundary artefact, it would vanish under PBC. Instead, the intercept becomes larger and more statistically significant.

The increase from $b_{\text{OBC}} \approx 0.145$ to $b_{\text{PBC}} \approx 1.17$ reflects the redistribution of long-range entanglement monogamy constraints around a closed loop, thickening the effective information gap and enhancing the emergent mass scale. The MI-derived Laplacian therefore describes a genuinely massive one-dimensional geometry whose mass gap persists in the thermodynamic limit and is robust to changes in topology.

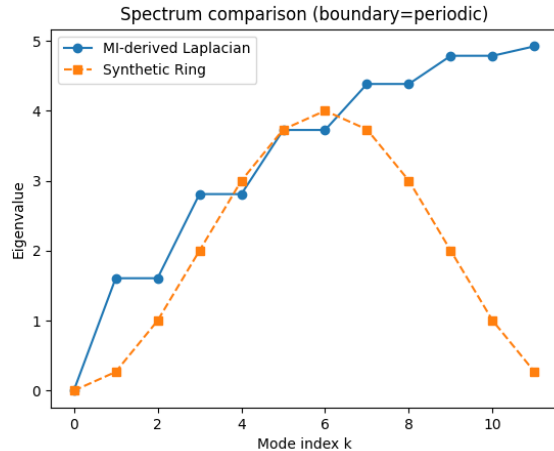


FIG. 12. Spectrum comparison between the MI-derived Laplacian and an ideal periodic ring Laplacian.

V. CONCLUSION

We have provided a comprehensive, statistically controlled numerical demonstration that differential-geometric structure can emerge from quantum entanglement alone in a simple spin chain. The persistence and enhancement of the λ_1 intercept under periodic boundaries shows that the emergent geometry contains a gen-

uine bulk mass scale. The pipeline developed here offers a concrete validation tool for information-theoretic approaches to emergent geometry and suggests several directions for follow-up, including extending to larger N using DMRG or MPS, applying the MI-to-Laplacian construction to higher-dimensional systems, and exploring connections to holographic and tensor-network models of emergent spacetime.

Appendix A: Computational pipeline

1. Exact diagonalization of the XXZ Hamiltonian for $N = 6, 8, 10, 12, 14, 16, 18$ at selected Δ .
2. Computation of the mutual-information matrix $I(i, j)$.
3. Construction of the normalized adjacency matrix and graph Laplacian.
4. Eigenvalue decomposition and extraction of eigenmodes.
5. Statistical analysis: AICc model selection, finite-size scaling, bootstrap analysis, synthetic benchmarking, boundary diagnostics, Luttinger-parameter sweep, and MI-decay exponent extraction.